

Exercise2: UNDERSTANDING DATA

Write Python programs to do the following operations:

1. Loading data from CSV file
2. Compute the basic statistics of given data - shape, no. of columns, mean
3. Splitting a data frame on values of categorical variables
4. Visualize data using Scatter plot

Dataset: brain_size.csv

Library: Pandas, matplotlib

- a) Loading data from CSV file
- b) Compute the basic statistics of given data - shape, no. of columns, mean
- c) Splitting a data frame on values of categorical variables
- d) Visualize data using Scatter plot

1. Loading data from CSV file

```
import pandas as pd
```

```
a=pd.read_csv("D:/data.csv")
```

```
print(a)
```

2. Compute the basic statistics of given data - shape, no. of columns, mean

```
import pandas as pd
```

```
a=pd.read_csv("D:/data.csv")
```

```
print('shape :',a.shape)
```

#no of columns

```
cols=len(a.axes[1])
```

```
print('no of columns:',cols)
```

#mean of data

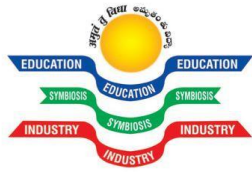
```
m=a["marks"].mean()
```

```
print('mean of marks:',m)
```

3. Splitting a data frame on values of categorical variables

```
import pandas as pd
```

```
a=pd.read_csv("D:\data.csv")
```



```
print("Before:")  
  
print(a)  
  
a_split=a['address'].str.split(' ',1)  
  
a['district']=a_split.str.get(0)  
  
a['state']=a_split.str.get(1)  
  
del(a['address'])  
  
print("After:")  
  
print(a)
```

4. Visualize data using Scatter plot

```
import pandas as pd  
  
import matplotlib.pyplot as plt  
  
a=pd.read_csv("D:\data.csv")  
  
print("Before",a)  
  
a_split=a['address'].str.split(', ',1)  
  
a['district']=a_split.str.get(0)  
  
a['state']=a_split.str.get(1)  
  
del(a['address'])  
  
print("After=",a)  
  
a.plot(kind='scatter',x='marks',y='rollno',c='red')  
  
plt.show()
```